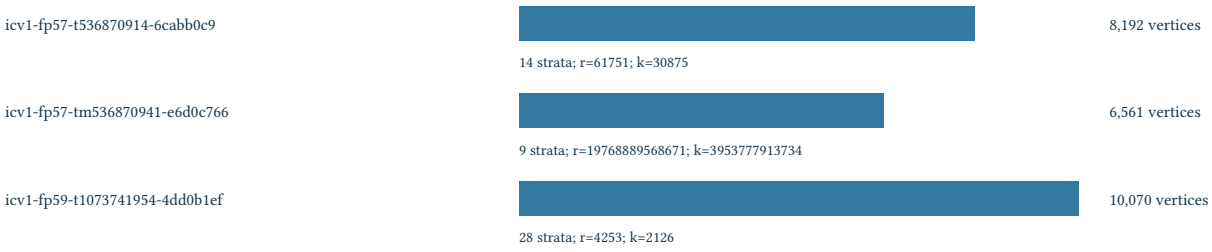
Full graph-wide IC: certified workload preparation

Run 20261017-inputs. This report contains **zero solver executions**.

Certified workload inputs across every stored vertex

Inputs only. Solver execution: NOT_RUN. No timing or relation-yield measurements.



Source: native evidence/inventory.json, run 20261017-inputs. Exact integer populations; no statistical uncertainty.

Source model	Vertices	Subgroup r	Embedding k
icv1-fp57-t536870914-6cabb0c9	8192	61751	30875
icv1-fp57-tm536870941-e6d0c766	6561	19768889568671	3953777913734
icv1-fp59-t1073741954-4dd0b1ef	10070	4253	2126

Observed: native preparation covered all 24,823 vertices and 51 conductor strata of the three additional complete graphs. Each input binds the exact model, conductor, largest-prime subgroup and deterministic nonzero generator. Curve membership and [r]G=O pass for every vertex. Original database SHA-256 values and every conductor population match the prior graph delivery.

Derived by native arithmetic: complete factorizations certify the selected prime subgroup and the exact multiplicative order of p modulo r. The finite-field destination has size p^k. No cheaper subgroup replaced the degree-3 class's r.

Scope: all-vertex direct EC relations and applicable finite-field IC, base grids, independent one-target workloads, matched references, full cold/online accounting, relation yields, rank, min/max/mean and explicit failures are required by the committed protocol. Input certification does not execute that matrix. Largest-prime-subgroup runs alone also do not measure full composite-order PH.

Still open: exact model/subgroup/generator representation registration, subgroup/quotient base validation, audited native control/pilot, a resumable full driver and provisioned execution/storage. Existing supersingular k=2 MOV uses BSGS/PH, not finite-field IC for these ordinary large-k destinations. This is a backend gap, not nonexistence of a mathematical reduction.

Evidence: evidence/inventory.json, inventory.stderr, implementation.sha256; per-vertex JSONL records and hashes accompany the inputs-only archive. Protocol commit 4253df8f2; native source commit e1b0e302e preceded execution. Linux x86_64, 4 CPU quota, 16 GiB RAM. Initial link exhausted temporary disk; stripped workspace relink succeeded. Failed outputs remain preserved.

No performance claim: solver runtime, operation distributions, relations and yield remain unknown. Independent implementation replay is not performed. IC/rho scoreboards retain their previous performance figures. Original A–E requirements, including full A/C/D/E generation, remain open. The draft backlog card is not a scheduled task; no solver job continues in the background.